

Ahmed Elbossily

CONTACT INFORMATION

Ahmed.Elbossily@leuphana.de

Website: AhmedElbossily.github.io

EDUCATION

Leuphana University Lüneburg
Institute for Production Technology and Systems

Lüneburg, Germany
2024–Present

Ph.D. Candidate

- Advisor: Prof. Dr.-Ing. Benjamin Klusemann
- Dissertation Topic: “GPU-Accelerated Smoothed Particle Hydrodynamics for Simulating Solid-State Manufacturing Processes”

**Arab Academy for Science, Technology, and
Maritime Transport (AASTMT)**

Cairo, Egypt
2014–2018

Master of Science (M.Sc.) in Computational Mechanics

- Grade: Excellent
- Advisor: Prof. Mostafa R. A. Atia

WORK EXPERIENCE

Research Assistant & Ph.D. Candidate

Leuphana University
2024–Present

- Conduct research within an ERC-funded project focused on GPU-accelerated solvers for manufacturing-process simulations.
- Develop and optimize high-performance C++/CUDA algorithms for large-scale particle-based simulations on NVIDIA GPUs.
- Profiled and analyzed CUDA developed solvers using NVIDIA Nsight Systems and Nsight Compute to identify performance bottlenecks, memory inefficiencies, and kernel-level optimization opportunities.
- Implement numerical models for plasticity, contact mechanics, friction, heat transfer, thermomechanical coupling, and deformation.
- Validate simulation results against experimental measurements of temperature, force, displacement, and material flow.
- Supervise graduation projects focused on GPU programming and parallel computing techniques.

Independent Consultant – Udacity

Udacity – Remote
2018–Present

- Delivered mentoring, live online sessions, project reviews, and technical support to improve learner outcomes.
- Supported corporate upskilling and accelerator programs, including C++ training for programmers from Stellantis.
- Mentored learners in C++, computer vision, machine learning, and artificial intelligence.
- Edit, review, and audit course materials, including the Autonomous Systems curriculum.

Engineering & Technical Leadership Roles

Egypt, Government Organization
2013–2024

Progressed through engineering, R&D, laboratory-management, advanced-systems, team lead, and technical-leadership positions.

- Developed software tools and engineering solutions for technical and operational challenges.
- Implemented and optimized GPU-accelerated algorithms, focusing on parallel execution and efficient memory access.
- Profiled CUDA applications using NVIDIA Nsight Systems and Nsight Compute.

- Planned and executed complex system tests, with a focus on control-system performance and ECU response.
- Analyzed test data, diagnosed system-level problems, and recommended technically feasible solutions.
- Led multidisciplinary engineering teams and prepared technical documentation, reports, and process instructions.

TEACHING EXPERIENCE

Leuphana University

2024–Present

- Introduction to GPU Programming
- C++ for Mechanical Applications
- Mathematics I–II (Exercises)

REFEREED PUBLICATIONS

1. **Ahmed Elbossily**, Z. Kallien, R. Chafle, K. A. Fraser, M. Afrasiabi, M. Bambach, and B. Klusemann, “GPU-accelerated meshfree computational framework for modeling the friction surfacing process,” *Computational Particle Mechanics*, Vol. 12, No. 5, pp. 3721–3745, 2025.
<https://doi.org/10.1007/s40571-025-01048-2>
2. **Ahmed Elbossily**, Z. Kallien, L. Rath, R. Chafle, M. Afrasiabi, and B. Klusemann, “Experimental–numerical investigation of force-controlled friction extrusion process via feedback-controlled simulation,” *Advances in Industrial and Manufacturing Engineering*, Vol. 12, 100184, 2026.
<https://doi.org/10.1016/j.aime.2026.100184>
3. **Ahmed Elbossily**, Z. Kallien, R. Chafle, K. A. Fraser, M. Afrasiabi, and B. Klusemann, “Numerical Investigation of Deposition Efficiency Influencing Factors in the Friction Surfacing Process,” *Key Engineering Materials*, Vol. 1050, pp. 1–7, April 2026. <https://doi.org/10.4028/p-4tzjoz>

TECHNICAL SKILLS

Programming: Applications:

C++, CUDA C/C++, Java, JavaScript, Python
NVIDIA Nsight Systems, Nsight Compute, GLM/C++,
Visual Studio, Android Studio, MATLAB, TensorFlow,
Inventor, SolidWorks, Abaqus, ParaView

Operating Systems:

Microsoft Windows and Linux

REFERENCES

Prof. Dr.-Ing. Benjamin Klusemann (Supervisor)

Professorship for Materials Mechanics
Institute of Production Engineering and Systems
Leuphana University Lüneburg
E-mail: benjamin.klusemann@leuphana.de